

COMP3311 Week 10

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Announcements

- Please complete the myExperience survey!
- Quiz 6 due Friday 2 August @11:59pm
- Just final exam left now!! 🙌
 - Monday 12 August (3-hour)
 - Allowed to bring in cheat sheet (double sided A4)
 - Make sure to watch live lectures or at least **look at the slides**.
e.g. Locking (in 22T3 Wk10) was not in tutorial questions, but was in the final exam 😬



Learning Objectives

01



Relational Algebra: Operators

02



Relational Algebra: Composition

03



Transaction Processing: Serialisability

04



Transaction Processing: Precedence Graphs

05



Transaction Processing: View Equivalence

01

Relational Algebra: Operators

01



Relational Algebra: Operators

- Like SQL but more mathematical
- **Selection:** $\text{sel}[\text{expr}](\text{rel})$
 - selects the rows in some relation that satisfies expression
- **Projection:** $\text{proj}[\text{a,b,c}](\text{rel})$
 - filters relation by its columns a, b, c
- **Rename:** $\text{rename}[\text{schema}](\text{rel})$
 - renames the columns of rel with the columns specified in schema

01

Relational Algebra: Operators

$\text{Sel}[B=1](R)$ is equivalent to

```
select *  
from R  
where B = 1;
```

Examples of selection:

R

A	B	C	D
a	1	x	4
b	2	y	5
c	4	z	4
d	8	x	5
e	1	y	4
f	2	x	5

$\text{Sel}[A=C](R)$

A	B	C	D

$\text{Sel}[B=1](R)$

A	B	C	D
a	1	x	4
e	1	y	4

$\text{Sel}[B \geq D](R)$

A	B	C	D
c	4	z	4
d	8	x	5

$\text{Sel}[A=b \text{ or } A=c](R)$

A	B	C	D
b	2	y	5
c	4	z	4

01

→ Relational Algebra: Operators

 $\text{Proj}[A, B, C](R)$

is equivalent to

```
select distinct A,B,C  
from R;
```

Examples of projection:

R

A	B	C	D
a	1	x	4
b	2	y	5
c	4	z	4
d	8	x	5
e	1	y	4
f	2	x	5

 $\text{Proj}[A,B,C](R)$

A	B	C
a	1	x
b	2	y
c	4	z
d	8	x
e	1	y
f	2	x

 $\text{Proj}[B,D](R)$

B	D
1	4
2	5
4	4
8	5

← Removes
duplicates !! $\text{Proj}[D](R)$

D
4
5

01



Relational Algebra: Operators

Rename[beerName](Proj[name](Beers))

is equivalent to

```
select  name as beerName  
from    Beers;
```


01

Relational Algebra: Operators

Union: Table1 Union Table2

Intersection: Table1 Intersection Table2

Difference: Table1 – Table2

- Returns tuples that exist in Table1 but DON'T exist in Table2
- NOT symmetric / NOT complement of Intersection
- i.e. Table1 – Table2 $\neq$ Table2 – Table1

$s1 = Sel [B = 1] (r1)$

A	B	C	D
a	1	x	4
e	1	y	4

$s1 - s2$

A	B	C	D
e	1	y	4

$s2 = Sel [C = x] (r1)$

A	B	C	D
a	1	x	4
d	8	x	5

$s2 - s1$

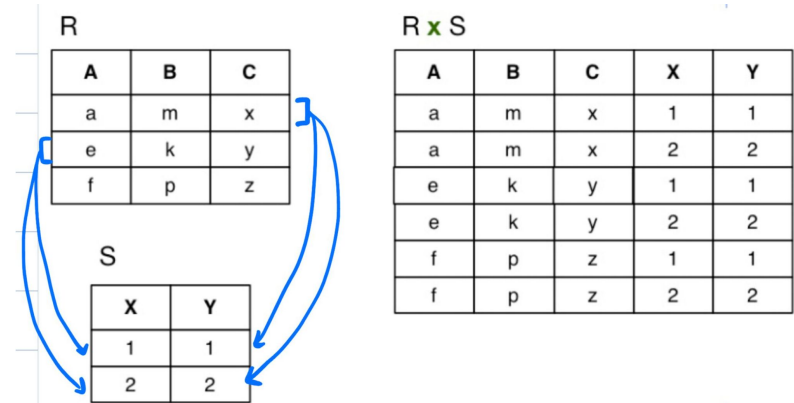
A	B	C	D
d	8	x	5

01

Relational Algebra: Operators

Product: Table1 x Table2

- Pairwise product between two tables
- for every row in Table1, attach every row in Table2
- T1: {a, b, c, d}, T2: {e, f}
- T1 x T2: {T1.a, T1.b, T1.c, T1.d, T2.e, T2.f}



01 → Relational Algebra: Operators

Division: Table1 Div Table2

- Returns all rows in T1 which have a match **for all** the rows in T2
- Captures the notion of “for-all”
 - i.e. returns all R that will satisfy T

R		S		T	R / T	S / T
A	B	A	B	B	A	A
4	x	4	x	x	4	4
4	y	4	y	y	5	
4	z	4	z			
5	x	5	x			
5	y	5	y			
5	z	5	z			

A		B	A/B
X	Y	Y	X
a	1	1	Null
b	2	2	
a	3	3	
d	4	4	

01



Relational Algebra: Operators

Natural join: Table1 join Table2

- Joins two tables on common attributes that are matched

Theta join: Table1 join [condition] Table2

- A natural join with a 'where' condition applied

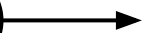
Outer join: Table1 [Full|Left|Right] Outer Join Table 2

- Left outer - all tuples from T1 included (T2 non matches set to null)
- Right outer - all tuples from T2 included (T1 non matches set to null)
- Full outer - both (left and right outer join applies)

02

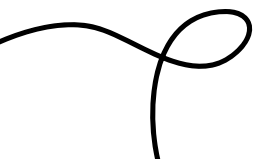
Relational Algebra: Composition

02



Relational Algebra: Composition

- **Composition:** Relational algebra expressions can be combined by using the result of one expression as the input of another expression
 - This allows us to compose simpler expressions which come together to answer arbitrarily complex queries.



02

Relational Algebra: Composition Questions

4. Consider the following two relations:

R		
A	B	C
a1	b1	c1
a1	b2	c2
a2	b1	c1

S	
B	C
b1	c1
b2	c2

Give the relation that results from each of the following relational algebra expressions on these relations:

- a. $R \text{ Div } S$
- b. $R \text{ Div } (\text{Sel}[B \neq b1](S))$
- c. $R \text{ Div } (\text{Sel}[B \neq b2](S))$
- d. $R \times S - (\text{Sel}[R.C=S.C](R \text{ Join}[B=B] S))$

02

Relational Algebra: Composition Answer

d. $R \times S - (\text{Sel}[R.C=S.C](R \text{ Join}[B=B] S))$

Answer:

$Tmp1 = R \text{ Join}[R.B=S.B] S$	=	<table><tr><th>R.A</th><th>R.B</th><th>R.C</th><th>S.B</th><th>S.C</th></tr><tr><td>a1</td><td>b1</td><td>c1</td><td>b1</td><td>c1</td></tr><tr><td>a1</td><td>b2</td><td>c2</td><td>b2</td><td>c2</td></tr><tr><td>a2</td><td>b1</td><td>c1</td><td>b1</td><td>c1</td></tr></table>	R.A	R.B	R.C	S.B	S.C	a1	b1	c1	b1	c1	a1	b2	c2	b2	c2	a2	b1	c1	b1	c1
R.A	R.B	R.C	S.B	S.C																		
a1	b1	c1	b1	c1																		
a1	b2	c2	b2	c2																		
a2	b1	c1	b1	c1																		

$\text{Tmp2} = \text{Sel}[\text{R.C}=\text{S.C}](\text{Tmp1})$	=	<table><tr><th>R.A</th><th>R.B</th><th>R.C</th><th>S.B</th><th>S.C</th></tr><tr><td>a1</td><td>b1</td><td>c1</td><td>b1</td><td>c1</td></tr><tr><td>a1</td><td>b2</td><td>c2</td><td>b2</td><td>c2</td></tr><tr><td>a2</td><td>b1</td><td>c1</td><td>b1</td><td>c1</td></tr></table>	R.A	R.B	R.C	S.B	S.C	a1	b1	c1	b1	c1	a1	b2	c2	b2	c2	a2	b1	c1	b1	c1
R.A	R.B	R.C	S.B	S.C																		
a1	b1	c1	b1	c1																		
a1	b2	c2	b2	c2																		
a2	b1	c1	b1	c1																		

$Tmp3 = R \times S$	=	<table><tr><th>R.A</th><th>R.B</th><th>R.C</th><th>S.B</th><th>S.C</th></tr><tr><td>a1</td><td>b1</td><td>c1</td><td>b1</td><td>c1</td></tr><tr><td>a1</td><td>b1</td><td>c1</td><td>b2</td><td>c2</td></tr><tr><td>a1</td><td>b2</td><td>c2</td><td>b1</td><td>c1</td></tr><tr><td>a1</td><td>b2</td><td>c2</td><td>b2</td><td>c2</td></tr><tr><td>a2</td><td>b1</td><td>c1</td><td>b1</td><td>c1</td></tr><tr><td>a2</td><td>b1</td><td>c1</td><td>b2</td><td>c2</td></tr></table>	R.A	R.B	R.C	S.B	S.C	a1	b1	c1	b1	c1	a1	b1	c1	b2	c2	a1	b2	c2	b1	c1	a1	b2	c2	b2	c2	a2	b1	c1	b1	c1	a2	b1	c1	b2	c2
R.A	R.B	R.C	S.B	S.C																																	
a1	b1	c1	b1	c1																																	
a1	b1	c1	b2	c2																																	
a1	b2	c2	b1	c1																																	
a1	b2	c2	b2	c2																																	
a2	b1	c1	b1	c1																																	
a2	b1	c1	b2	c2																																	

<i>Tmp3</i> - <i>Tmp2</i>	=	<table><tr><th>R.A</th><th>R.B</th><th>R.C</th><th>S.B</th><th>S.C</th></tr><tr><td>a1</td><td>b1</td><td>c1</td><td>b2</td><td>c2</td></tr><tr><td>a1</td><td>b2</td><td>c2</td><td>b1</td><td>c1</td></tr><tr><td>a2</td><td>b1</td><td>c1</td><td>b2</td><td>c2</td></tr></table>	R.A	R.B	R.C	S.B	S.C	a1	b1	c1	b2	c2	a1	b2	c2	b1	c1	a2	b1	c1	b2	c2
R.A	R.B	R.C	S.B	S.C																		
a1	b1	c1	b2	c2																		
a1	b2	c2	b1	c1																		
a2	b1	c1	b2	c2																		

03

Transaction Processing: Serialisability

03

Transaction Processing: Serialisability

- **Transaction:**

- A set of atomic (single operation) actions that a user performs
e.g. a read or a write

- **Serial execution:**

- No overlap between 2 or more transactions → or

```
T1: R(X) W(X) R(Y) W(Y)
T2:                R(X) W(X)
```

- **Serialisable schedule:**

- Given some transactions, I can create a schedule such that it would in effect be similar to a serial execution of them.

```
T1:                R(X) W(X) R(Y) W(Y)
T2: R(X) W(X)
```

03



Transaction Processing: Serialisability

To maintain the integrity of data, transactions must be:

- **Atomic:**
 - Each operation is a single, unsplittable unit
 - Either each operation fully succeeds, or fully fails and rolls back fully
- **Consistent:**
 - Each operation takes a valid database and results in a valid database
- **Isolated:**
 - If several operations are performed together, it should be possible to perform them separately to the same effect
- **Durable:**
 - Resistant to data corruption from events such as power loss

03

Transaction Processing: Serialisability

Conflict serialisable schedule:

- A schedule that guarantees the **outcome of concurrent transactions is the same as the serial order of the same transactions**.
 - This means that the concurrent schedule doesn't violate any data dependencies or constraints that would be enforced by the serial order
- It's stricter and easier to enforce by **not allowing conflicts in the order of operations**
- How to check for conflict serialisable?
 - Conflict serialisable if there are **no cycles in precedence graph**

03

→ Transaction Processing: Serialisability

- An operation is **non-conflicting** if swapping the order doesn't affect the result
 - Basically can't swap anything with a write operation

Consider two transactions T_1 and T_2 acting on data item X .

Possible orders for read/write operations by T_1 and T_2 :

T_1 first	T_2 first	Equiv?
$R_1(X) R_2(X)$	$R_2(X) R_1(X)$	yes
$R_1(X) W_2(X)$	$W_2(X) R_1(X)$	no
$W_1(X) R_2(X)$	$R_2(X) W_1(X)$	no
$W_1(X) W_2(X)$	$W_2(X) W_1(X)$	no

If T_1 and T_2 act on different data items, result is always equivalent.

03

Transaction Processing: Serialisability

View serialisable schedule:

- It assesses whether the **final outcome of a schedule is equivalent in view to a serial schedule.**
 - It is a schedule that allows all read operations to “see” the correct version of data.
- This ensures a consistent and predictable database
- It's broader and ensures the same final state but **allows different order of operations as long as final outcomes are the same.**
- The check is only performed if schedule is not conflict-serialisable
 - **As a conflict serialisable schedule will also be view serialisable.**
- (See later slides on view equivalence and how we can determine this)

04

Transaction Processing: Precedence Graph

04

Transaction Processing: Precedence Graph

- If something needs to **write** or **read** before something else, then draw an edge
- If there **is a cycle**, then **NOT conflict serialisable**

04

Transaction Processing: Precedence Graph

```
T1:  R(X) W(X)           R(Y)
T2:           R(X) W(Y)
```

T1 → T2 (draw an edge because T1 has to finish writing X before T2 reads X)

T2 → T1 (draw an edge because T2 has to finish writing Y before T1 reads Y)

→ There is a cycle → NOT conflict serialisable

04

Transaction Processing: Precedence Graph Question

11. Draw the precedence graph for the following schedule (where C means "commit"):

T1:	R(A) W(Z)	C
T2:	R(B) W(Y)	C
T3:	W(A)	W(B) C

12. [based on Ramakrishnan, ex.17.2]

Consider the following incomplete schedule S:

T1:	R(X) R(Y) W(X)	W(X)
T2:	R(Y)	R(Y)
T3:	W(Y)	

- Determine (by using a precedence graph) whether the schedule is conflict-serializable
- Modify S to create a complete schedule that is conflict-serializable

05

Transaction Processing: View Equivalence

05

Transaction Processing: View Equivalence

View equivalence properties checklist:

1. If a transaction performs the first read on the original, it must also perform the first read on the serialised schedule
2. If a transaction performs the final write on the original, it must also perform the final write on the serialised schedule
3. Any order between a **read and write** must be the same in both schedules

How to check for view serialisable?

- **View serialisable if there is a view equivalent schedule in all possible serial schedules**
- If no possible view equivalent schedule then it's not view serialisable
- A conflict serialisable schedule is also view serialisable

05

Transaction Processing: View Equivalence Question

13. [based on Ramakrishnan, ex.17.3]

For each of the following schedules, state whether it is conflict-serializable and/or view-serializable. If you cannot decide whether a schedule belongs to either class, explain briefly. The actions are listed in the order they are scheduled, and prefixed with the transaction name.

- a. T1:R(X) T2:R(X) T1:W(X) T2:W(X)
- b. T1:W(X) T2:R(Y) T1:R(Y) T2:R(X)
- c. T1:R(X) T2:R(Y) T3:W(X) T2:R(X) T1:R(Y)
- d. T1:R(X) T1:R(Y) T1:W(X) T2:R(Y) T3:W(Y) T1:W(X) T2:R(Y)
- e. T1:R(X) T2:W(X) T1:W(X) T3:W(X)



AND THAT'S IT!

All the best for your final exam!!!

**Hope to see you all around
sometime 🖐️**

